

Hypothesis Testing (Explained in the Simplest Way)

Imagine you are a teacher.

One day, the principal says:

"The average marks of Class A are 70."

You don't know whether this statement is true or false.

So, you decide to check by taking marks from some students.

This process is called **Hypothesis Testing**.

What is Hypothesis Testing?

Simple Definition

Hypothesis Testing is a method of checking whether a claim is true or false using sample data.

It is similar to an investigation.

Someone makes a claim.

↓

You collect data.

↓

You decide whether the claim is correct.

Real-Life Example 1

Your friend says

"This restaurant has the best pizza."

Can you believe it immediately?

No.

You eat the pizza.

Then decide.

Hypothesis Testing works in the same way.

Real-Life Example 2

A company says

"Our new mobile battery lasts 20 hours."

To verify this:

- Test 100 mobiles.
- Record battery life.
- Decide whether the claim is correct.

This is Hypothesis Testing.

Why Do We Need Hypothesis Testing?

Suppose there are **10 lakh students**.

Can you ask every student their marks?

 No.

Instead:

- Select 100 students.
- Calculate their average.
- Make a decision about all students.

This saves:

- Time
 - Money
 - Effort
-

Important Terms

1. Population

Population means **everyone or everything** you want to study.

Example:

- All students in a school
- All employees in a company
- All customers of Amazon

Example

Population = 10,000 Students

2. Sample

A sample is **a small group taken from the population.**

Example

Population = 10,000 Students

↓

Sample = 100 Students

Instead of checking everyone, we check only a few.

What is a Hypothesis?

A hypothesis is simply **a claim or statement.**

Example:

Average salary is ₹50,000.

or

Students study 5 hours daily.

These are hypotheses.

Types of Hypothesis

There are only **two hypotheses.**

1. Null Hypothesis (H_0)

Null means

Nothing has changed.

or

The claim is true.

Example

Company says

Average Salary = ₹50,000

Then

H_0

Average Salary = ₹50,000

We start by assuming this claim is true.

2. Alternative Hypothesis (H_1)

Alternative means

Something is different.

Example

H_1

Average Salary $\neq$ ₹50,000

Meaning

The company's claim is wrong.

Easy Example

Teacher says

Average Marks = 70

Then

H_0

Average Marks = 70

Alternative

H_1

Average Marks $\neq$ 70

Think Like a Judge

Imagine a judge in a court.

The judge first assumes

The accused is innocent.

Only if there is strong evidence does the judge decide otherwise.

Hypothesis testing follows the same idea.

We first assume **H_0 is true**, then look for evidence to reject it.

How Hypothesis Testing Works

Suppose

Company claims

Average Salary = ₹50,000

Step 1

Write the claim

H_0

Salary = ₹50,000

Step 2

Collect sample

Take salary of

100 Employees

Step 3

Calculate average

Suppose average comes

₹35,000

Step 4

Ask yourself

Is ₹35,000 close to ₹50,000?

Probably not.

The claim looks wrong.

Step 5

Decision

Reject the company's claim.

Another Example

A coaching institute says

Every student scores above 80.

You check 50 students.

Average comes

52

Will you believe the coaching institute?

No.

Reject the claim.

P-value (Very Easy Explanation)

Students often fear the p-value, but it is simple.

Think Like This

Suppose you throw a coin.

It lands

Head

Head

Head

Head

Head

Head

Head

Head

Head

Head

10 Heads in a row!

Now ask yourself

Is this normal?

Probably not.

Maybe the coin is not fair.

The **p-value** tells us **how surprising our result is if the original claim (H_0) is true.**

Simple Rule

If

$P\text{-value} < 0.05$

Result is surprising.

Reject H_0 .

If

$P\text{-value} \geq 0.05$

Result is normal.

Do not reject H_0 .

Easy Way to Remember

P-value	Decision
Less than 0.05	Reject H_0
Greater than or equal to 0.05	Fail to Reject H_0

Example

Suppose

$P\text{-value} = 0.02$

Since

$0.02 < 0.05$

Reject H_0 .

Suppose

P-value = 0.40

Since

$0.40 > 0.05$

Do not reject H_0 .

Why 0.05?

It means

We accept only a **5% chance of making a wrong decision**.

It is an internationally accepted standard in many fields, though sometimes 0.01 or 0.10 is used depending on the situation.

Type I Error (Easy Example)

Suppose

A healthy person is diagnosed as sick.

Reality

Healthy

Doctor says

Sick

Wrong decision.

This is **Type I Error** (False Positive).

Type II Error (Easy Example)

Reality

Person is Sick

Doctor says

Healthy

Wrong again.

This is **Type II Error** (False Negative).

Where is Hypothesis Testing Used in Machine Learning?

1. Feature Selection

Suppose we have these features:

- Age
- Salary
- Hair Color

We test whether each feature has a meaningful relationship with the target.

If Hair Color is not useful, we remove it.

2. Compare Two Models

Model A Accuracy

90%

Model B Accuracy

92%

Is the 2% improvement real or just due to chance?

Hypothesis testing helps answer that.

3. A/B Testing

A company creates two websites.

Website A

Old

Website B

New

They measure which website gets more purchases.

4. Medical Research

Doctors test

Old Medicine

vs

New Medicine

to determine if the new medicine is actually better.

Simple Python Example

Suppose the average marks should be **70**.

We test whether our class average is different.

```
from scipy.stats import ttest_1samp
```

```
marks = [72, 75, 70, 68, 74, 73, 71]
```

```
t_stat, p_value = ttest_1samp(marks, 70)
```

```
print("T-statistic:", t_stat)
```

```
print("P-value:", p_value)
```

If the output is:

P-value = 0.03

Since

$0.03 < 0.05$

We reject the null hypothesis.

This suggests the class average is significantly different from 70.

Flow Chart

Someone makes a claim



Write H_0 and H_1



Collect Sample Data



Calculate Test Statistic & P-value



Is P-value < 0.05 ?

